

UNIVERSITY OF OKARA
Department of Computer Science

ARTIFICIAL INTELLIGENCE

Lecture 01: Introduction to Artificial Intelligence

Comprehensive Student Handout
BS Computer Science — 3rd Semester

Instructor: Ghulam Ali

Course: Artificial Intelligence | Program: BS Computer Science | Semester: 3rd

How to use this handout

Use this document to prepare before the lecture, follow along during class, revise afterward, and study for exams. It goes deeper than the lecture slides — the slides are for classroom delivery; this handout is your full reference on every concept introduced in Lecture 01.

Table of Contents

Learning Objectives

1. What Is Artificial Intelligence?
 2. Scope of Artificial Intelligence
 3. AI and Human Intelligence
 4. A Brief History of Artificial Intelligence
 5. AI vs Machine Learning vs Deep Learning vs Generative AI
 6. Machine Learning
 7. Deep Learning
 8. Generative AI
 9. Foundation Models
 10. Large Language Models
 11. Transformers and Attention
 12. AI Agents
 13. Major Branches of AI
 14. Computer Vision
 15. Natural Language Processing
 16. Robotics
 17. Reinforcement Learning
 18. Real-World Applications of AI
 19. AI in Computer Science
 20. Modern AI Technology Landscape
 21. AI Skills for Computer Science Students
 22. In-Class Activities Reference
- Quick-Revision Summary & Key Terms

Learning Objectives

After studying this handout and attending Lecture 01, students should be able to:

- Define Artificial Intelligence (AI) in both formal and student-friendly terms.
- Explain the scope of AI by describing its major capabilities (learning, reasoning, perception, language, planning, decision-making, and generation).
- Compare human intelligence and artificial intelligence across multiple dimensions, and correctly evaluate the claim that AI is "as intelligent as humans."
- Describe the major milestones and eras in the history of AI, from the 1950s to the present.
- Distinguish between Artificial Intelligence, Machine Learning, Deep Learning, and Generative AI, and correctly classify a given system into the appropriate category or categories.
- Explain, at a conceptual level, how Machine Learning, Deep Learning, Foundation Models, and Generative AI work.
- Identify the major branches of AI (NLP, Computer Vision, Robotics, Reinforcement Learning, and others) and give an example application for each.
- Identify real-world applications of AI across multiple industries.
- Explain how AI integrates with other areas of Computer Science (software engineering, cybersecurity, databases, networks, cloud computing, HCI, and more).
- Recognize the opportunities and limitations of modern AI systems, and avoid common misconceptions about what AI can and cannot do.
- Identify the foundational skills (programming, data handling, ML/DL, LLMs, RAG, AI agents) a Computer Science student should build across the degree to work with AI.

1. What Is Artificial Intelligence?

1.1 Definition of Artificial Intelligence

There is no single universally agreed definition of Artificial Intelligence, because "intelligence" itself is difficult to define precisely, even for humans. It is useful to look at a few academically credible definitions before settling on a working definition for this course.

John McCarthy (who coined the term "Artificial Intelligence" in 1955): described AI as "the science and engineering of making intelligent machines."

Stuart Russell and Peter Norvig (authors of the standard AI textbook): frame AI as the design of agents that perceive their environment and take actions that maximize their chance of achieving a goal — sometimes measured against human performance, sometimes against an ideal rational standard.

A widely used working definition: AI is the branch of computer science concerned with building systems that can perform tasks which, when performed by humans, are understood to require intelligence.

Synthesizing these views, the working definition used throughout this course (matching the lecture slide) is:

Artificial Intelligence is the field of computing concerned with building systems that can perform tasks requiring capabilities commonly associated with human intelligence.

It is worth unpacking each word in this definition:

- **Artificial** — not naturally occurring; created and engineered by humans, as opposed to intelligence that arises biologically.
- **Intelligence** — the capacity to acquire and apply knowledge or skill to achieve goals; in this context, capabilities such as learning, reasoning, and problem-solving.
- **Machine / system** — any computational artifact — software, hardware, or a combination — that carries out the intelligent behavior. This does not have to be a physical robot.
- **Intelligent system** — a system that exhibits one or more intelligent capabilities (see Section 1.2) rather than following only fixed, hand-written rules.
- **Intelligent behavior** — behavior that would normally be described as requiring thought, judgment, or skill if performed by a human — for example, recognizing a face, translating a sentence, or planning a route.

1.2 What Does AI Try to Achieve?

AI is not one single algorithm or technique — it is a broad field that covers many different capabilities. The lecture slide lists eight of the most important ones. Each is explained below with a practical, everyday example.

Capability	What It Means	Everyday Example
Learning	Improving performance on a task using data or experience, rather than being explicitly programmed for every case.	A spam filter gets better at catching junk mail the more labeled examples it sees.
Reasoning	Drawing conclusions or making inferences from available information.	A medical decision-support tool inferring a likely diagnosis from a set of symptoms.

Capability	What It Means	Everyday Example
Problem Solving	Finding a sequence of steps or a solution to reach a defined goal.	A chess engine searching for the best next move.
Perception	Interpreting raw sensory input such as images, sound, or sensor data.	A phone unlocking itself using face recognition.
Language Understanding	Extracting meaning from human language, written or spoken.	A voice assistant understanding "set an alarm for 7 AM."
Decision Making	Choosing the best action among alternatives, often under uncertainty.	A navigation app choosing the fastest route given current traffic.
Planning	Arranging a sequence of actions in advance to achieve a goal.	A warehouse robot planning an efficient path to collect items.
Generation	Producing new content — text, images, audio, code — rather than just classifying or predicting.	An app automatically writing a caption for a photo.

A useful way to remember this: AI is not one thing that a machine either "has" or "doesn't have." It is a set of capabilities, and a real system usually combines several of them. For example, a voice assistant combines language understanding, decision making, and generation (the spoken reply) in a single interaction.

1.3 Is AI Really "Intelligent"?

This is one of the most important conceptual points in the entire course, and it is easy to misunderstand. It is important to be precise here rather than making sweeping claims in either direction.

- **Intelligent behavior** refers to the output a system produces — an outcome that looks intelligent when observed from the outside. A system can produce intelligent behavior in a narrow context without possessing anything like genuine understanding.
- **Human intelligence** is general, flexible, grounded in lived experience, embodied, and accompanied by consciousness and common sense — properties that are far broader than what any current AI system has.
- **Task-specific (narrow) intelligence** describes almost all AI systems in active use today. A system trained to detect spam, translate text, or recognize faces performs very well at that one task but has no general understanding beyond it, and typically cannot transfer its skill to an unrelated task without retraining.
- **Artificial General Intelligence (AGI)** is a hypothetical future system that would match or exceed human-level intelligence across essentially any intellectual task, with the flexibility and generality that humans have. AGI does not exist today, and there is no scientific consensus on when — or whether — it will be achieved.

Important — do not overclaim

Current AI systems, including today's most advanced Large Language Models, are highly capable at specific classes of tasks but are not equivalent to human-level general intelligence. Referring to any existing system as "AGI" or as "thinking" in the human sense is inaccurate and is a common misconception this course will repeatedly correct.

2. Scope of Artificial Intelligence

Because AI covers so many capabilities, it is best understood as an umbrella field rather than a single technique. The lecture's "Big Picture" slide maps AI as a central hub connecting many capability areas. A simplified version of that map is shown below.

Conceptual Map of AI

AI

- Learning — acquiring patterns or skills from data or experience
- Reasoning — drawing logical conclusions from facts or rules
- Perception — interpreting visual, auditory, or sensor input (Vision, Speech)
- Language — understanding and producing human language
- Planning — sequencing actions to reach a goal
- Decision Making — choosing among alternatives, often under uncertainty
- Prediction — estimating future or unknown values from known data
- Generation — producing new content
- Interaction — engaging with humans or other systems (dialogue, interfaces)
- Robotics — combining perception, planning, and action in the physical world

Each of these is not a separate, isolated product — they are building blocks that combine inside real systems. A single self-driving car, for instance, draws on perception (seeing the road), planning (choosing a path), and decision making (reacting to a pedestrian) all at once. A voice assistant draws on speech perception, language understanding, decision making, and generation (the spoken reply).

Why this matters

Students who think of AI only as "chatbots" or "robots" are seeing two narrow slices of a much larger field. Later sections of this handout (Sections 13–17) explore each of these capability areas as a formal "branch" of AI with its own techniques and applications.

3. AI and Human Intelligence

Comparing human intelligence with artificial intelligence helps clarify both what AI is good at and where it still falls short. The table below expands the comparison shown in the lecture across the dimensions most relevant to computer science students.

Dimension	Human Intelligence	Current AI Systems
Learning source	Learns from lived, embodied experience — often from very few examples.	Learns from large datasets; typically needs many examples (though some modern models generalize from fewer).
Memory	Associative, reconstructive, and selective; can forget or distort details.	Can store and precisely recall vast amounts of information without fatigue.
Reasoning	Flexible; can reason about entirely novel situations using common sense.	Strong at pattern-based inference within familiar domains; weaker at genuinely novel, out-of-distribution situations.
Perception	Multi-sensory, contextual, and robust to noise and ambiguity.	Very strong within trained domains (e.g., image or speech recognition) but can fail in unexpected ways outside training conditions.
Adaptability	Adapts readily to unfamiliar situations, sometimes after a single experience.	Generally requires retraining or fine-tuning to adapt to a genuinely new task or domain.
Common sense	Rich, implicit understanding of how the everyday world works.	Limited; a well-known weak point of current AI, even in advanced language models.
Creativity	Originates novel ideas from lived experience, emotion, and cultural context.	Can recombine and generate content in novel-looking ways based on learned patterns, but does not originate ideas from lived experience.
Speed / scale	Limited processing speed; cannot easily parallelize across millions of cases.	Can process, compare, and act on enormous volumes of data extremely quickly.
Energy efficiency	The human brain runs on roughly the power of a light bulb.	Training and running large AI models can require substantial computational power and energy.
Generalization	Transfers understanding across very different domains and tasks.	Often narrow; performs well on the task it was trained for, less reliably outside it.
Emotional understanding	Genuine emotional experience and empathy grounded in consciousness.	Can recognize patterns associated with emotional language or expression, without genuine emotional experience.

The single most important takeaway from this comparison

Most AI systems today are specialized / task-oriented — not equivalent to general human intelligence. Avoid simplistic statements such as "AI is smarter than humans." It is more accurate to say that AI can outperform humans in some narrowly defined tasks (for example, playing Go, or scanning millions of transactions for fraud) while remaining limited in other areas that humans handle naturally, such as common-sense reasoning about an unfamiliar physical situation.

A helpful illustration: a child who touches a hot stove once learns "hot stove = danger" permanently, from a single experience, and can generalize that lesson to hot pans, hot irons, and hot pavement. An image classification system, by contrast, typically needs thousands of labeled example images before it can reliably recognize a new category — and even then, it may not generalize the concept of "hot" beyond the visual patterns it was trained on.

4. A Brief History of Artificial Intelligence

AI is often assumed to be a brand-new field that appeared alongside ChatGPT. In reality, it has more than 70 years of continuous development, organized into several recognizable eras. Understanding this history helps explain why the field looks the way it does today, and why certain ideas (such as neural networks) took decades to become practical.

Early Foundations (before 1950)

Long before electronic computers existed, mathematicians and logicians laid the theoretical groundwork AI would later depend on: formal mathematical logic (rules for valid reasoning), the theory of computation (what it means for a problem to be "computable," formalized by figures such as Alan Turing), and very early ideas about artificial neurons, proposed in the 1940s as a simplified mathematical model of how a biological neuron might work. These ideas made it possible, in principle, to imagine machines that could reason or learn — long before the necessary computing power existed.

The 1950s: AI Becomes a Field

- 1950 — Alan Turing publishes his famous question, "Can machines think?", and proposes what is now known as the Turing Test as a practical way to evaluate machine intelligence through conversation.
- 1956 — The Dartmouth Workshop brings together early researchers and formally introduces the term "Artificial Intelligence," marking the birth of AI as a recognized academic field.

1960s–1970s: Symbolic AI

Early AI research focused on symbolic AI — representing knowledge as symbols and logical rules, and using search algorithms to find solutions (for example, proving mathematical theorems or solving puzzles). This period also produced early natural language systems and early robotics research, though systems of this era were narrow and struggled outside tightly controlled problems.

1980s: Expert Systems

The 1980s were dominated by expert systems: programs that encoded the knowledge of human experts (e.g., doctors or engineers) as large sets of hand-written if-then rules, used to give advice or make decisions in a narrow domain such as medical diagnosis. Expert systems demonstrated real commercial value but were expensive to build and maintain, and could not learn or adapt automatically — every rule had to be written by hand.

AI Winters

An "AI winter" refers to a period of reduced funding and interest in AI research, typically following a period of inflated expectations that the technology of the time could not deliver on. The field experienced more than one such winter (notably in the mid-1970s and late 1980s/early 1990s), as the limitations of symbolic AI and expert systems became apparent and results fell short of earlier promises.

Lesson from the AI winters

Overpromising relative to what the underlying technology can actually deliver tends to produce a backlash. This is a recurring pattern worth remembering as you evaluate claims about today's AI systems.

1990s: Statistical Approaches

Research shifted away from purely hand-written rules toward statistical machine learning — systems that learn patterns directly from data rather than requiring every rule to be specified in advance. A landmark event of this decade was IBM's Deep Blue defeating world chess champion Garry Kasparov in 1997 — an early, highly visible demonstration of a narrow AI system outperforming a human expert at one specific task.

2000s: More Data, More Computing Power

Steadily increasing computational power, along with the growth of large digital datasets (partly driven by the rise of the internet), made more sophisticated machine learning methods practical at a larger scale than before.

2010s: The Deep Learning Era

- 2012 — AlexNet, a deep neural network, dramatically outperforms prior approaches on the ImageNet image-recognition competition — widely regarded as the breakthrough that triggered the modern deep learning boom.
- 2016 — AlphaGo defeats world-class Go player Lee Sedol. Unlike Deep Blue's hand-tuned search strategy, AlphaGo learned largely through self-play using reinforcement learning, a qualitatively different and more general approach.

This decade also produced major advances in speech recognition and computer vision, driven by deep neural networks.

2017 Onward: The Transformer Era

In 2017, researchers introduced the Transformer architecture, built around a mechanism called attention. This architectural shift, combined with large-scale pretraining on massive datasets, became the technical foundation for nearly every major AI breakthrough that followed, including modern Large Language Models. (Section 11 explains attention and Transformers in more depth.)

2020s: Foundation Models and Generative AI

The current decade has been defined by foundation models — very large models pretrained once on broad data and then adapted to many tasks — Large Language Models (LLMs), Generative AI, multimodal AI (systems that combine text, images, audio, and more), AI coding assistants, AI agents, and growing use of AI for scientific discovery.

Year / Period	Milestone	Why It Matters
1940s	Early neural / computational ideas	Provided the mathematical groundwork later used by neural networks.
1950	Turing asks "Can machines think?"	Introduced a concrete, testable way to think about machine intelligence.
1956	Dartmouth Workshop — AI named as a field	Marks the formal founding of AI as an academic discipline.
1960s–70s	Symbolic AI & problem solving	Established search and logic-based reasoning as early AI techniques.
1980s	Expert Systems	First commercially significant AI applications; exposed the limits of hand-written rules.
1997	Deep Blue defeats Kasparov	High-profile proof that narrow AI could beat a human expert at one specific task.

Year / Period	Milestone	Why It Matters
2012	AlexNet — deep learning breakthrough	Triggered the modern deep learning boom in computer vision and beyond.
2016	AlphaGo defeats Lee Sedol	Showed learning-through-self-play (reinforcement learning) surpassing hand-tuned search.
2017	Transformer architecture	Became the technical foundation of virtually all modern large-scale AI models.
2020s	Foundation models, LLMs, Generative AI	Enabled general-purpose models adaptable to a huge range of downstream tasks.
2025–26	Rise of AI agents & AI-for-science	AI systems increasingly plan and act using tools, not just answer questions.

Common misconception

"AI is brand new — it started with ChatGPT." In reality, AI is the product of more than 70 years of continuous research, spanning at least three broad eras: (1) the symbolic / rule-based era (1950s–1980s), (2) the statistical machine-learning era (1990s–2010s), and (3) the deep learning and generative era (2012–present).

Note: dates above are drawn from commonly cited AI history sources (e.g., the Stanford AI Index and ACM/AAAI historical summaries) and should be treated as a representative, not exhaustive, timeline.

5. AI vs. Machine Learning vs. Deep Learning vs. Generative AI

This is one of the most important — and most frequently confused — distinctions in the entire course. Getting it right now will make every later lecture easier to follow.

5.1 The Conceptual Relationship

At a high level, these four terms sit at different "depths" of specificity:

- Artificial Intelligence (AI) — the broad field of intelligent computational systems. It is the largest circle: everything else in this list is a part of AI.
- Machine Learning (ML) — a major approach within AI, in which systems learn patterns from data rather than following only hand-written rules. ML sits inside AI.
- Deep Learning (DL) — a specific approach within ML that uses multilayer neural networks to learn patterns. DL sits inside ML.
- Generative AI — a capability (creating new content) that, in today's mainstream tools, is commonly built on deep learning and large foundation models.

Important nuance — read carefully

Generative AI is NOT always a strict subset of Deep Learning in every possible context. Some generative approaches have historically existed outside deep learning (for example, simpler statistical generative models). What is true is that modern, mainstream Generative AI systems — such as today's image generators and chatbots — commonly rely on deep learning and large foundation models. Avoid the oversimplified claim that "Generative AI is simply another layer below Deep Learning": the term Generative AI refers primarily to what a system does (generate content), not to one specific underlying technique.

5.2 Side-by-Side Comparison

Concept	Definition	Main Idea	Typical Data	Examples
AI	The broad field of building systems that perform tasks requiring human-like capabilities.	Umbrella field covering many capabilities and techniques.	Any — structured, unstructured, symbolic, or none at all (e.g., rule-based systems).	Expert systems, search algorithms, ML systems, robots.
ML	Systems that learn patterns from data instead of following only explicit rules.	Train a model on data so it can make predictions or decisions on new data.	Structured / labeled or unlabeled datasets.	Spam filters, recommendation engines, fraud detectors.
DL	Machine learning using multilayer ("deep") neural networks.	Learn layered, increasingly abstract representations directly from raw data.	Large volumes of raw data — images, audio, text.	Face recognition, speech recognition, modern LLMs.
Generative AI	Systems that create new content rather than only classify or predict.	Given a prompt/input, produce new text, images, audio, code, or other content.	Massive, broad pretraining data; commonly deep-learning based today.	ChatGPT-style assistants, image generators, AI coding tools.

A few concrete systems, placed at their different "depths":

- A spam filter that uses statistical rules learned from labeled emails — classic Machine Learning (not necessarily Deep Learning).
- A face-recognition system built on a convolutional neural network — Machine Learning AND Deep Learning.
- A chatbot that writes essays, built on a large pretrained language model — Machine Learning, Deep Learning, AND Generative AI.

A self-driving car's object detector (which classifies "pedestrian" vs. "not pedestrian") sits in ML/DL territory — it classifies, it does not generate new content. A chatbot that writes original essays sits in the Generative AI territory, built on top of DL. Being able to place a system correctly is a key skill this course will keep reinforcing.

6. Machine Learning

Machine Learning (ML) is the branch of AI in which a system improves its performance on a task by learning patterns from data, rather than being explicitly programmed with every rule in advance.

6.1 The Machine Learning Pipeline

A typical ML system follows this pipeline:

Data → Learning Algorithm → Model → Prediction / Decision

Each stage has a specific meaning:

Data: the raw examples the system learns from (e.g., thousands of past emails).

Learning algorithm (training): the process by which the system analyzes the data and adjusts itself to capture useful patterns.

Model: the result of training — a mathematical object that has "learned" patterns from the data and can be applied to new, unseen inputs.

Features: the measurable input characteristics the model uses to make a prediction (e.g., words present in an email, sender address, number of links).

Labels: the known correct answers attached to training examples in supervised learning (e.g., "spam" or "not spam").

Inference / Prediction: using a trained model to produce an output for a new input it has not seen before.

Common misconception

Students often assume "the model" is hand-written code containing explicit if-then logic. In reality, the model's behavior comes from patterns learned FROM data during training — not from rules a programmer typed out by hand.

Worked example from the lecture:

Email → [Learning Algorithm] → Spam / Not Spam

A spam filter is trained on thousands of emails that have already been labeled "spam" or "not spam" by humans. The learning algorithm finds statistical patterns that distinguish the two categories (certain words, sender patterns, formatting) and produces a model. That model can then classify brand-new, unlabeled incoming emails.

6.2 The Three Major Learning Paradigms

Paradigm	How It Learns	Example
Supervised Learning	Learns from labeled examples — each training input comes with a known correct output.	Spam detection: emails labeled "spam" / "not spam" in advance.
Unsupervised Learning	Finds structure or patterns in unlabeled data — there is no "correct answer" given in advance.	Grouping customers into segments based on purchasing behavior, with no predefined categories.

Paradigm	How It Learns	Example
Reinforcement Learning	Learns through trial, error, and reward feedback while interacting with an environment.	An agent learning to play a game by receiving rewards for good moves and penalties for bad ones.

Self-check example: If you wanted to group students into study groups based on similar quiz scores, without any correct grouping given in advance — is that supervised or unsupervised learning? (Answer: unsupervised — there is no predefined "correct" grouping; the system discovers structure in the data on its own.)

Scope note

This section is only an introduction to Machine Learning. A full treatment of ML algorithms, evaluation methods, and mathematics will come in later, dedicated lectures of this course.

7. Deep Learning

Deep Learning (DL) is a branch of Machine Learning that uses neural networks with multiple layers to learn useful representations directly from data.

7.1 Artificial Neurons and Neural Networks

A neural network is loosely inspired by the structure of biological neurons, though it is a simplified mathematical model rather than a biological simulation. It is organized into layers:

Input layer: receives the raw data (e.g., the pixel values of an image).

Hidden layer(s): intermediate layers that each transform the data into a more abstract representation. "Deep" refers to the number of layers stacked between input and output — not to any notion of intelligence "depth."

Output layer: produces the final result (e.g., a predicted label, such as "cat").

Input → Hidden Layer → Hidden Layer → Output

A helpful mental picture, using image recognition as an example: early layers might learn to detect simple features like edges; middle layers combine those into shapes; later layers combine shapes into whole objects such as "cat" or "car." This progressive build-up of abstraction is called representation learning — the network learns, on its own, which features of the data are useful, rather than a human hand-engineering those features.

Common misconception

"Deep" learning does not mean the network thinks "deeply" or has deep understanding. It refers strictly to having many stacked layers of computation.

7.2 Major Deep Learning Architectures (Names Only)

This course will not cover the mathematics of these architectures in Lecture 01 — only what kind of problem each one is historically strong at, as a preview for later lectures.

Architecture	Full Name	Strong For
CNN	Convolutional Neural Network	Images — detecting spatial patterns such as edges, shapes, and objects.
RNN	Recurrent Neural Network	Sequences — data where order matters, such as time series or (historically) text.
Transformer	Attention-based architecture	Sequences at large scale — the architecture that powers modern Large Language Models (see Section 11).

Typical Deep Learning application areas include image recognition, speech recognition, and language understanding — all tasks that involve finding complex patterns in large, high-dimensional data (pixels, audio waveforms, or word sequences).

8. Generative AI

8.1 What Is Generative AI?

Generative AI refers to systems that take a prompt or input and produce genuinely new content, as opposed to traditional predictive models that only assign a label or a number to an input.

This is a fundamentally different task shape from traditional (predictive/discriminative) AI:

Traditional AI: Input → Model → Prediction (e.g., a label or a number) Generative AI: Prompt / Input → Model → Generated Content (something new)

For example, a traditional image classifier takes a photo and outputs a label such as "dog." A generative image model takes a text prompt such as "a golden retriever running on a beach at sunset" and produces a brand-new picture that did not exist before.

8.2 Types of Content Generative AI Can Produce

- Text — essays, summaries, code comments, conversational replies
- Images — illustrations, photorealistic pictures, design concepts
- Audio — speech synthesis, sound effects, voice cloning
- Video — short clips, animations
- Code — functions, scripts, entire small programs
- Music / 3D content — original compositions, 3D models and scenes

Common misconception

Many students think Generative AI means only text chatbots such as ChatGPT. In reality, the output types above show that text, images, audio, video, code, and music are all active areas of Generative AI.

8.3 The Role of Prompts and Context

A prompt is the instruction or input a user provides to a generative model. The quality, specificity, and context given in a prompt strongly influence the quality and relevance of the generated output — this is why "prompting" is treated as a skill in its own right (see Section 21). Context — earlier parts of a conversation, an uploaded document, or examples given in the prompt — shapes how the model interprets what is being asked.

9. Foundation Models

A foundation model is a model trained once, on massive and broad data, and then adapted to many different downstream tasks — rather than a brand-new model being built from scratch for every single application.

Pretraining → Foundation Model → Adaptation → Applications

Large-scale pretraining: training a model on an enormous, broad dataset (text, images, code, and more) before it is adapted to any specific task.

General capabilities: the broad, flexible abilities the model acquires during pretraining, useful across many different tasks rather than just one.

Adaptation: the process of tailoring a pretrained foundation model to a specific task, typically through fine-tuning (further training on task-specific data) or prompting (giving the model instructions and examples at use-time, without changing its underlying parameters).

Downstream task: a specific, real-world application the foundation model is adapted to, such as summarization, customer support, or code completion.

The lecture's summary phrase captures the whole idea concisely:

"Train once at large scale → adapt for many tasks."

Categories of foundation models include:

- Language Models — pretrained primarily on text
- Vision Models — pretrained primarily on images
- Vision-Language Models — pretrained on paired image and text data
- Multimodal Models — able to process and/or generate combinations of text, images, audio, and more

Why foundation models matter

The same underlying pretrained language model can be adapted to power a customer-support chatbot, a code assistant, and a document summarizer — all without training three completely separate models from zero. This reuse is a major reason foundation models have changed the economics and practice of modern AI: it is far cheaper to adapt one large pretrained model than to build a new model for every task.

10. Large Language Models

A Large Language Model (LLM) is a type of foundation model trained on massive amounts of text, primarily to predict what word (or word-fragment) comes next in a sequence. Despite this deceptively simple training objective, LLMs at large scale can perform an enormous range of language tasks.

10.1 Core Concepts

Language model: a model that captures statistical patterns of language, enabling it to estimate how likely a given sequence of words is, and to generate plausible continuations.

Tokens: the small units (often word-pieces or sub-words) that text is broken into before being processed by the model. "Tokenization" is the step of splitting text into tokens.

Context window: the maximum amount of text (measured in tokens) a model can consider at once when generating a response.

Next-token prediction: the core training task of most LLMs: given the text so far, predict the most likely next token.

Pretraining: the initial, large-scale training phase on broad text data (see Section 9).

Instruction following: additional training that teaches a pretrained model to follow user instructions helpfully and safely, rather than just continuing text in a generic way.

Fine-tuning: further training of a pretrained model on a smaller, more specific dataset to specialize its behavior for a particular task or domain.

Prompting: crafting the input text given to an LLM to obtain a desired kind of response, without changing the model's underlying parameters.

Hallucination: when a model generates plausible-sounding but factually incorrect or fabricated content, stated with apparent confidence.

10.2 A Critical Clarification

Important — do not overclaim

An LLM does not automatically "search the internet" simply because it is an LLM. A base LLM generates responses using patterns learned during training, up to its training cutoff. Some AI products connect an LLM to external tools, such as a web search tool or a document retrieval system (see RAG in Section 20) — but that capability comes from the surrounding system design, not from being an LLM by itself.

This distinction matters because it directly explains hallucination: since an LLM predicts plausible next tokens rather than looking up verified facts by default, it can produce confident-sounding statements that are not true. Being aware of hallucination is an essential, practical skill for anyone using LLM-based tools, including in academic work.

11. Transformers and Attention

The Transformer architecture, introduced in 2017, is the technical foundation behind nearly every major AI system released since — including modern LLMs, many vision models, and multimodal systems. Understanding why it became so important, at a conceptual level, is worthwhile even before studying its mathematics in a later course.

11.1 The Processing Pipeline

Text → Tokens → Embeddings → Attention → Transformer Layers → Output

Tokens: the text broken into small units, as introduced in Section 10.

Embeddings: numeric representations of tokens — each token is converted into a list of numbers (a vector) that captures aspects of its meaning, so the model can perform mathematical operations on language.

Attention: a mechanism that lets the model weigh every token against every other token in the context, to figure out which parts of the input are most relevant to each other.

Transformer layers: stacked processing layers, each refining the model's understanding using attention and other operations, before producing a final output.

11.2 Attention, in Plain Language

Consider this sentence, used in the lecture as a worked example:

"The animal didn't cross the road because it was tired."

A human reader immediately understands that "it" refers to "the animal," not "the road" — but a computer has no built-in notion of this unless it can relate words to each other using context. Attention is the mechanism that lets the model weigh every word against every other word to figure out that "it" refers to "the animal," not "the road." This is not a hard-coded grammar rule — it emerges from patterns learned during training on huge amounts of text.

Common misconception

Transformers do not "understand" language the way humans do. Attention is a statistical weighting mechanism over learned representations — extremely powerful and useful, but fundamentally different from human comprehension, which is grounded in lived experience and consciousness.

This lecture deliberately avoids Transformer mathematics. The goal at this stage is only the core idea: context shapes meaning, word by word — and attention is the mechanism that lets a model use that context.

12. AI Agents

AI agents represent one of the newest and fastest-growing areas of applied AI, and a preview of a concept this course will return to in more depth later.

12.1 What Is an AI Agent?

An AI agent is a system that perceives its environment, reasons or plans about what to do, and takes actions toward a goal — potentially using external tools, and observing the results of its actions to decide what to do next.

An AI agent can be designed to:

- Receive information from its environment or from a user
- Reason about that information
- Plan a sequence of steps toward a goal
- Use tools (e.g., a search engine, a code interpreter, a database)
- Take actions in a digital or physical environment
- Observe the results of those actions
- Continue toward the goal, repeating the cycle as needed

Goal → Observe → Reason → Plan → Act → Observe Result → Repeat

12.2 Examples of AI Agents

- A research assistant that searches for information, reads sources, and compiles a summary
- A coding agent that writes, tests, and debugs code toward a specified goal
- A customer-service agent that answers questions and can take actions such as issuing a refund
- A scheduling assistant that checks calendars and books meetings on a user's behalf

Important clarification

The word "agent" can refer to systems with very different levels of autonomy — from a simple system that suggests one action for a human to approve, to a more autonomous system that can chain together many actions with little human oversight. Not every AI system described as an "agent" has the same level of independence.

13. Major Branches of AI

AI is not one branch — it is a family of interconnected sub-fields. Several of these were already named in earlier sections (ML, DL, Generative AI); the table below lists the major branches together with a definition, the main problem each addresses, and a representative example and application.

Branch	Definition	Main Problem	Example / Application
Machine Learning	Learning patterns from data.	Making predictions or decisions without explicit rules.	Recommendation systems, fraud detection.
Deep Learning	ML using multilayer neural networks.	Learning complex patterns from raw, high-dimensional data.	Image and speech recognition.
Natural Language Processing	Processing, understanding, and generating human language.	Bridging human language and computation.	Chatbots, translation, summarization.
Computer Vision	Interpreting and understanding visual information.	Extracting meaning from images/video.	Face recognition, medical imaging.
Robotics	Combining perception, decision-making, and physical action.	Operating intelligently in the physical world.	Warehouse robots, autonomous vehicles.
Expert Systems	Encoding human expert knowledge as rules.	Automating decisions in a narrow domain.	Early medical diagnosis systems.
Knowledge Representation	Structuring knowledge so machines can use it.	Representing facts and relationships formally.	Knowledge graphs, ontologies.
Reasoning	Drawing logical conclusions from known facts.	Inference and deduction.	Automated theorem proving.
Planning	Sequencing actions to reach a goal.	Deciding what to do, and in what order.	Robot path planning, logistics.
Speech AI	Processing and generating spoken language.	Converting between speech and text/meaning.	Voice assistants, transcription.
Reinforcement Learning	Learning via trial, error, and reward feedback.	Learning optimal behavior through interaction.	Game-playing agents, robotics control.
Generative AI	Creating new content from a prompt.	Producing novel text, images, audio, code.	AI writing and image-generation tools.
Multimodal AI	Combining multiple data types (text, image, audio).	Reasoning across more than one kind of input/output.	Systems that describe images in words.
AI Agents	Systems that plan and act using tools toward a goal.	Autonomous or semi-autonomous task completion.	Coding agents, research assistants.
Explainable AI	Making AI decisions understandable to humans.	Opening the "black box" of complex models.	Explaining why a loan application was declined.

Branch	Definition	Main Problem	Example / Application
Trustworthy / Responsible AI	Ensuring AI systems are safe, fair, and accountable.	Reducing bias, harm, and misuse.	Fairness audits of hiring algorithms.

Common misconception

These branches are not separate and unrelated. Real systems often combine several branches at once — for example, a hospital triage assistant might combine NLP (reading doctors' notes), Computer Vision (reading medical scans), and Explainable AI (justifying its recommendation to clinicians).

14. Computer Vision

Computer Vision is the branch of AI concerned with enabling computers to interpret and understand visual information — extracting meaning from pixels, in much the same way NLP extracts meaning from text.

Image → Vision Model → Understanding

14.1 Core Tasks

- Image classification — assigning a label to an entire image (e.g., "cat")
- Object detection — locating and identifying multiple objects within an image
- Image segmentation — labeling every pixel of an image according to what object it belongs to
- Face recognition — identifying or verifying a specific individual from a face
- Optical Character Recognition (OCR) — converting text within an image into machine-readable text
- Medical imaging analysis — detecting patterns of clinical interest in scans such as X-rays or MRIs

14.2 Applications

- Face recognition — unlocking phones, identity verification
- Medical imaging — flagging suspicious regions on an X-ray for a radiologist to review
- Autonomous vehicles — detecting lanes, pedestrians, and other vehicles
- Object detection — retail inventory tracking, security systems
- Quality inspection — spotting manufacturing defects on a production line
- Satellite imagery / agriculture — monitoring crop health from aerial or satellite images

Common misconception

"Seeing" for a computer is not the same as human vision. Computer Vision performs pattern recognition over pixel data — it does not have genuine visual experience the way a human observer does.

15. Natural Language Processing

Natural Language Processing (NLP) is the branch of AI concerned with enabling computers to process, understand, and generate human language — both written and spoken.

Human Language → AI (NLP Model) → Understanding / Generation

NLP covers both understanding language (e.g., sentiment analysis) and generating it (e.g., chatbot replies, summaries). LLMs (Section 10) are today's dominant NLP technology, but NLP as a field is broader than LLMs alone.

15.1 Core Applications

- Text classification — sorting text into categories (e.g., topic, urgency)
- Sentiment analysis — determining whether text expresses a positive, negative, or neutral opinion
- Machine translation — converting text from one language to another
- Summarization — condensing long text into a shorter version that preserves key meaning
- Question answering — producing a direct answer to a natural-language question
- Chatbots — conversational agents that respond to user input
- Speech-related applications — connecting spoken and written language (see also Section 17's Sense–Think–Plan–Act loop for how this feeds into agents)

Common misconception

NLP is not "only chatbots." Translation, search, and summarization are equally central applications — chatbots are simply one of the most visible.

16. Robotics

Robotics combines perception, decision-making, and physical action, connecting Sensors, an AI / Decision System, and Actuators (the physical components that carry out an action, such as motors or grippers).

Sense → Think → Plan → Act

This Sense–Think–Plan–Act loop repeats continuously while a robot operates: sensors gather information about the environment ("Sense"), the AI system interprets that information and reasons about it ("Think"), a course of action is chosen ("Plan"), and the robot's actuators carry it out ("Act") — after which the cycle repeats using updated sensor data.

16.1 Example Domains

- Industrial robots — assembly lines, precision manufacturing
- Drones — aerial inspection, delivery, mapping
- Autonomous vehicles — self-driving cars and trucks
- Service robots — hospitality, cleaning, assistance
- Warehouse robots — sensing shelf locations via cameras/sensors, planning the most efficient path, and acting by moving and picking items

Common misconception

"Robot" does not always mean a humanoid machine. Robots take many forms — drones, warehouse arms, and self-driving cars are all robots in the AI sense of the term.

A basic definition worth remembering: an AI agent (introduced formally in Section 12) is a system that perceives its environment, reasons/plans, and takes actions toward a goal — robotics is the branch of AI most concerned with agents that act in the physical world.

17. Reinforcement Learning

Reinforcement Learning (RL) is best understood by analogy: think of it like training a player to win a game through practice. An agent takes an action in an environment, receives a reward or penalty, and uses that feedback to improve future decisions — repeating this loop many times.

Agent → Action → Environment → Reward / Penalty → (back to) Agent → Learning

Agent: the learner or decision-maker.

Environment: everything the agent interacts with and receives feedback from.

Action: a choice the agent makes at a given moment.

Reward / penalty: feedback signal indicating how good or bad an action turned out to be.

Policy: the agent's strategy for choosing actions (name only — the details are covered in a later, dedicated lecture).

Common misconception

RL does not need labeled correct answers the way supervised learning does. RL learns from trial-and-error feedback (reward signals) rather than being told the correct action in advance.

17.1 Applications

- Game playing — e.g., AlphaGo, introduced in the AI history timeline (Section 4), learned to play Go largely through self-play using reinforcement learning
- Robotics — learning physical control policies through trial and error
- Resource optimization — learning efficient strategies for allocating limited resources
- Autonomous decisions — systems that improve their decision-making through repeated feedback

A relatable analogy: a video game player who levels up through repeated practice — trying strategies, losing, adjusting, and eventually improving — is following essentially the same feedback loop an RL agent follows, just without a formal mathematical reward signal.

18. Real-World Applications of AI

AI adoption is broad across every industry — not limited to technology companies. The subsections below follow a consistent pattern: Problem → AI Approach → Example → Benefit → Potential Limitation.

Healthcare

- **Problem:** Detecting disease patterns in medical data is slow and requires scarce specialist time.
- **AI approach:** Deep learning models trained on labeled medical images.
- **Example:** A model trained on thousands of labeled X-rays flags anomalies for a radiologist to review.
- **Benefit:** Faster triage; can catch patterns humans might miss.
- **Potential limitation:** Model errors could delay correct diagnosis if used without expert oversight.

Finance

- **Problem:** Fraudulent transactions must be caught quickly among huge transaction volumes.
- **AI approach:** ML classifiers trained on transaction patterns.
- **Example:** Real-time fraud and risk detection in banking and mobile payment apps.
- **Benefit:** Faster fraud detection at a scale no manual team could match.
- **Potential limitation:** False positives can block legitimate transactions.

Education

- **Problem:** Every student learns differently, but personalized attention doesn't scale.
- **AI approach:** Adaptive learning systems, NLP-based tutoring tools.
- **Example:** Personalized tutoring that adjusts to a student's pace and gaps.
- **Benefit:** More individualized learning support.
- **Potential limitation:** Risk of over-reliance reducing independent problem-solving skills.

Cybersecurity

- **Problem:** Novel attacks are hard to catch with fixed rule sets.
- **AI approach:** Anomaly detection using ML on network traffic/logs.
- **Example:** Threat detection systems that flag unusual network behavior.
- **Benefit:** Faster detection of previously unseen attack patterns.
- **Potential limitation:** Attackers can adapt to evade learned patterns (adversarial risk).

Agriculture

- **Problem:** Monitoring large farmland for crop health is labor-intensive.
- **AI approach:** Computer vision on satellite/drone imagery.
- **Example:** Crop monitoring for early detection of disease or stress.
- **Benefit:** Earlier intervention, better yields.
- **Potential limitation:** Accuracy depends on imagery quality and local calibration.

Transportation

- **Problem:** Traffic and routing decisions are complex and dynamic.
- **AI approach:** ML-based prediction and optimization.
- **Example:** Route optimization in navigation apps.
- **Benefit:** Reduced travel time and congestion.
- **Potential limitation:** Predictions can degrade under unusual/rare conditions.

Manufacturing

- **Problem:** Unexpected equipment failure is costly.
- **AI approach:** ML models predicting equipment wear from sensor data.
- **Example:** Predictive maintenance that flags machines likely to fail soon.
- **Benefit:** Reduced downtime and repair cost.
- **Potential limitation:** Requires reliable sensor data and historical failure records.

Entertainment

- **Problem:** Helping users find relevant content among vast catalogs.
- **AI approach:** Recommendation algorithms based on user behavior.
- **Example:** Content recommendation on streaming platforms.
- **Benefit:** More relevant content discovery.
- **Potential limitation:** Can create narrow "filter bubbles" if not designed carefully.

Software Engineering

- **Problem:** Writing and reviewing code is time-consuming.
- **AI approach:** LLM-based coding assistants.
- **Example:** AI code assistance for writing, completing, and reviewing code.
- **Benefit:** Faster development; can catch some errors early.
- **Potential limitation:** Generated code still needs human review for correctness and security.

Additional application areas mentioned in the lecture

Smart cities and scientific research are two further domains where AI is increasingly applied — for example, AI-optimized traffic and utility systems in smart cities, and AI-assisted hypothesis generation or data analysis in scientific research. These will be explored further in later lectures.

19. AI in Computer Science

This section is particularly important for BS Computer Science students: AI is no longer a separate elective topic — it is a capability increasingly integrated into nearly every CS discipline you will study.

CS Area	How AI Integrates
Software Engineering	Code generation & review — AI coding assistants suggest, complete, and review code.

CS Area	How AI Integrates
Cybersecurity	Intrusion & anomaly detection — ML models flag unusual network or system behavior.
Data Science	Predictive analytics — ML/DL models extract patterns and forecasts from data.
Databases	Query optimization — AI-based optimizers improve how queries are executed.
Computer Networks	Traffic anomaly prediction — ML models detect unusual network traffic patterns.
Cloud Computing	Auto-scaling & resource management — AI decides when to allocate or release compute resources.
Internet of Things (IoT)	Smart sensor decision-making — on-device or edge AI makes local decisions from sensor data.
Human-Computer Interaction (HCI)	Adaptive user interfaces — interfaces that adjust based on learned user behavior.
Robotics	Autonomous control systems — AI plans and controls physical robot behavior.

Concrete examples: modern databases increasingly use AI-based query optimizers to improve performance, and cloud platforms use AI to make auto-scaling decisions in real time.

Common misconception

"Only AI/ML majors need to learn AI." In reality, AI now touches networks, databases, cybersecurity, and HCI too — foundational AI literacy is increasingly a baseline expectation across Computer Science roles, not a specialization reserved for a few.

20. Modern AI Technology Landscape

This section provides a conceptual map connecting everything covered so far into one end-to-end picture. Its purpose is to give you a mental map for the rest of the degree, not to serve as an advanced technical tutorial.

Data ↓ *Machine Learning* ↓ *Deep Learning* ↓ *Foundation Models* ↓ *LLMs / Vision Models* ↓ *Generative AI* ↓ *RAG* ↓ *AI Agents* ↓ *AI Applications*

Raw data flows "up" through ML and DL techniques, into foundation models, which are then specialized (e.g., as LLMs or vision models). On top of that sits Generative AI, then applied technologies such as Retrieval-Augmented Generation (RAG) and AI agents, which finally power real, deployed AI applications.

RAG (Retrieval-Augmented Generation): a technique that gives a generative model access to external documents or knowledge before it answers — helping ground its output in specific, up-to-date, or private information rather than relying only on what it learned during pretraining.

Example of the full stack in action: a company might use a foundation LLM, add Retrieval-Augmented Generation to ground it in their own internal documents, and wrap it in an AI agent to build an automated customer-support tool.

Common misconception

These layers are not competing alternatives where a newer layer replaces an older one. They build on each other — a modern AI agent, for example, still depends on the foundation model, and the foundation model still depends on deep learning and data underneath it.

Reminder: Today: just the map. Later courses in this degree will provide the depth behind each layer.

21. AI Skills for Computer Science Students

This section answers a question every student in this course will eventually ask: "What should I actually learn to work with AI?" The roadmap below is intended to unfold across your entire BS degree — not to be mastered all at once.

Reassurance

This is a learning roadmap for your whole degree — not something to master all at once. Mastery builds gradually, course by course, project by project.

21.1 The Progressive Roadmap

Level	Skill Area	What It Involves
1	Programming	The foundation for everything else — writing, reading, and debugging code.
2	Data Handling	Collecting, cleaning, and organizing data so it is usable for ML.
3	Machine Learning	Understanding core algorithms, training, and evaluation.
4	Deep Learning	Neural networks and representation learning for complex data.
5	Generative AI	Working with models that create new content.
6	LLMs and APIs	Using large language models programmatically via APIs.
7	RAG	Grounding model outputs in external knowledge sources.
8	AI Agents	Building systems that plan and act using tools.
9	Model Evaluation and Responsible AI	Assessing quality, fairness, safety, and accountability of AI systems.

Responsible AI: a preview concept covering fairness, safety, and accountability in AI systems — ensuring AI is built and used in ways that avoid harm and bias. This will be revisited in more depth later in the course.

21.2 Tools and Skills You Will Build Hands-On Experience With

- **Python** — General-purpose programming language, the dominant language for AI/ML work.
- **Git / GitHub** — Version control and collaborative software development.
- **SQL** — Querying and managing structured data in databases.

- **PyTorch / Scikit-learn** — Core libraries for building and training ML/DL models.
- **Hugging Face** — A widely used platform and library ecosystem for sharing and using pretrained models.
- **LLM APIs** — Programmatic interfaces for calling large language models from code.
- **Prompting** — The skill of crafting effective inputs to get useful outputs from generative models.
- **Data Analysis** — Exploring, cleaning, and interpreting datasets.
- **Model Evaluation** — Measuring how well a model performs, and where it fails.

Industry job postings increasingly list Python, Git, SQL, and "familiarity with LLM APIs" as baseline skills even for roles that are not AI-specialist positions — reinforcing that this roadmap has value well beyond students who plan to specialize in AI research.

22. In-Class Activities Reference

Lecture 01 includes several interactive activities. This reference summarizes each one so you can prepare in advance or revisit what was discussed.

Warm-Up: What Comes to Your Mind When You Hear "AI"?

A quick warm-up surfacing prior knowledge before AI is formally defined. You are asked to write down one example of AI you personally used that day (e.g., ChatGPT, self-driving cars, face recognition, recommendations, robots, voice assistants, image generation, fraud detection).

Quick Challenge: Is This AI?

For each of eight everyday items, decide: AI? Not AI? Depends? Items include a calculator, spam filter, face recognition, maps route planning, Netflix recommendation, ChatGPT, thermostat, and autonomous vehicle.

The key test for this activity: does the system exhibit learning, reasoning, or pattern-based decision-making, or does it only follow fixed, pre-programmed rules?

- Calculator — Not AI (fixed rules, no learning).
- Thermostat — usually Not AI (simple feedback control), though "smart" learning thermostats can be AI.
- Spam filter, face recognition, maps routing, Netflix recommendation, ChatGPT, autonomous vehicle — AI (all use learned models or optimization).

Evolution Activity: A Brief History of AI (Timeline Challenge)

Working in pairs, students try to guess the decade of a milestone before it is revealed — reinforcing the timeline in Section 4 of this handout.

Key Distinction Activity: Classification Challenge

For a named tool or system (e.g., Netflix recommender, self-driving perception, image generator, spam filter), identify which layer(s) of the AI / ML / DL / Generative AI relationship (Section 5) it belongs to.

Generative AI: Think-Pair-Share

Discuss with a partner which Generative AI tools you have already used, then share with the class.

Field Map Activity: Branch Identification

For a named product (self-driving car, hospital chatbot, warehouse robot), identify which 2–3 branches of AI (Section 13) it likely uses.

Applications Activity: Real-World Application Challenge

In groups by industry, propose one additional AI use case for your assigned industry beyond what was shown in Section 18.

Capstone Group Activity: Can You Design an AI Solution?

Working in one of six groups, each assigned a real-world problem — detect fake news, detect cyber attacks, predict student performance, recognize plant diseases, build a university chatbot, or detect road accidents — answer the following five questions and present your answers to the class:

- 1. What is the problem?
- 2. What data might be needed?
- 3. What AI capability is required?
- 4. Which AI branch could solve it?
- 5. What could be the benefit?

Tip for approaching this activity: work through the five questions in order. It is a common mistake to jump straight to a proposed solution without first identifying what data would actually be needed.

Closing Reflection

Write down one real-world problem you would personally like to solve using AI. This is a useful exercise to revisit as the course progresses — your answer may become more specific and technically grounded as you learn more.

Quick-Revision Summary

Use this page for last-minute revision before a quiz or exam.

- AI is the field of computing concerned with building systems that perform tasks requiring capabilities commonly associated with human intelligence — it is a broad umbrella, not one algorithm.
- Most AI systems today are narrow / task-specific, not equivalent to general human intelligence (AGI does not currently exist).
- AI has evolved through recognizable eras: symbolic AI (1950s–80s) → expert systems (1980s) → AI winters → statistical ML (1990s–2010s) → deep learning (2012–) → Transformers (2017–) → foundation models & Generative AI (2020s–).
- AI $\supset$ Machine Learning $\supset$ Deep Learning, while Generative AI is a capability (creating content) commonly, but not exclusively, built on Deep Learning and foundation models today.
- ML pipeline: Data → Learning Algorithm → Model → Prediction. Three paradigms: Supervised, Unsupervised, Reinforcement Learning.
- Deep Learning uses multilayer neural networks; "deep" refers to the number of layers, not intelligence. CNN → images; RNN → sequences; Transformer → large-scale sequences (modern LLMs).
- Generative AI: Prompt/Input → Model → Generated Content (text, images, audio, video, code, music) — distinct from traditional predictive AI.
- Foundation models: Pretraining → Foundation Model → Adaptation → Applications — "train once, adapt for many tasks."
- LLMs predict the next token; key risks/concepts: hallucination, context window, prompting, fine-tuning. An LLM does not automatically search the internet.
- Transformers use attention to weigh every token against every other token in context — the foundation of modern LLMs.
- AI agents: Goal → Observe → Reason → Plan → Act → Observe Result → Repeat — systems that use tools and take actions toward a goal.
- Major branches of AI include ML, DL, NLP, Computer Vision, Robotics, Reasoning, Planning, Speech AI, Reinforcement Learning, Generative AI, Multimodal AI, AI Agents, Explainable AI, and Trustworthy/Responsible AI.
- AI applications span nearly every industry: healthcare, finance, education, cybersecurity, agriculture, transportation, manufacturing, entertainment, and software engineering.
- AI increasingly integrates across every CS discipline: software engineering, cybersecurity, data science, databases, networks, cloud computing, IoT, HCI, and robotics.
- Skills roadmap for the degree: Programming → Data Handling → ML → DL → Generative AI → LLMs/APIs → RAG → AI Agents → Model Evaluation & Responsible AI.

Key Terms Glossary

Term	Definition
Artificial Intelligence (AI)	The field of computing concerned with building systems that perform tasks requiring human-like capabilities.
Machine Learning (ML)	Systems that learn patterns from data rather than following only explicit, hand-written rules.
Deep Learning (DL)	Machine learning using multilayer ("deep") neural networks.
Generative AI	Systems that create new content (text, images, audio, etc.) from a prompt.
Foundation Model	A model pretrained at large scale and adapted to many downstream tasks.
Large Language Model (LLM)	A foundation model trained primarily on text, using next-token prediction.
Transformer	The attention-based neural network architecture behind most modern large-scale AI models.
Attention	A mechanism allowing a model to weigh the relevance of different parts of the input to each other.
Token	A small unit (word or sub-word) that text is broken into for processing.
Hallucination	Plausible-sounding but false or fabricated output generated by an AI model.
Prompt	The input or instruction given to a generative AI model.
Fine-tuning	Further training of a pretrained model on task-specific data.
Supervised Learning	Learning from labeled examples (input paired with a known correct output).
Unsupervised Learning	Finding structure or patterns in unlabeled data.
Reinforcement Learning	Learning through trial, error, and reward/penalty feedback.
Convolutional Neural Network (CNN)	A deep learning architecture especially strong for image data.
Recurrent Neural Network (RNN)	A deep learning architecture especially strong for sequential data.
AI Agent	A system that perceives, reasons, plans, and takes actions toward a goal, often using tools.
Retrieval-Augmented Generation (RAG)	Giving a generative model access to external documents before it answers, to ground its output.
Artificial General Intelligence (AGI)	A hypothetical future AI with human-level general intelligence across virtually any task; does not currently exist.
Narrow / Task-Specific AI	AI that performs well on a specific task but lacks general intelligence beyond it.
Expert System	A rule-based AI program that encodes human expert knowledge for a narrow domain.
AI Winter	A historical period of reduced AI funding and interest following unmet expectations.

Looking ahead

"AI is not just about ChatGPT or robots. AI is a way of building computational systems that can learn, reason, perceive, generate, and make decisions." The next lecture will go deeper into one of the branches introduced here — most likely Machine Learning fundamentals. Your AI journey starts today.